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12 February 1987
SUPERSEDING
MIL-G-3859D
26 October 1972

MILITARY SPECIFICATION

# GREASE GUNS, HAND-OPERATED, LEVER, PUSH AND SCREW TYPE

# GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the general requirements for various type hand-operated grease guns for delivering light or heavy bodied lubricants to grease fittings.

1.2 Classification. The grease guns shall be of the following types, sizes, classes and outlets as specified (see 6.2):

Type I - Lever Operated (MIL-G-3859/1)

Size 14 14 ounce
Size 18 18 ounce
Size 80 80 ounce

Type II - Push Operated (MIL-G-3859/2)

Size 03 3 ounce
Class A - Straight body
Class B - Angle body.

Size 10 8 to 12 ounce
Size 28 28 ounce

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Screw Operated (MIL-G-3859/3

Size 08 8 ounce
Outlet A
Outlet B

Type IV - Pistol Grip (MIL-G-3859/4)

Size 03 3 ounce
Size 14 14 ounce

1.3 Part number. Specification part numbers for items described in this specification will be identified as shown in the applicable specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-P-40 | - Packaging and Packing of Hand Tools. |

MILITARY

| | |
|---|---|
| MIL-F-3541 | - Fittings, Lubrication. |
| MIL-G-3859/1 | - Grease Guns, Hand Operated, Lever, Push and Screw Type; Type I, Lever Operated. |
| MIL-G-3859/2 | - Grease Guns, Hand Operated, Lever, Push and Screw Type; Type II, Push Operated. |
| MIL-G-3859/3 | - Grease Guns, Hand Operated, Lever, Push and Screw Type; Type III, Screw Operated. |
| MIL-G-3859/4 | - Grease Guns, Hand Operated, Lever, Push and Screw Type; Type IV, Pistol Grip. |
| MIL-L-4387 | - Lubrication Dispensing Equipment Accessories. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-H-13444 | - Hose, Rubber: Fuel and Oil. |
| MIL-C-22587 | - Cartridge, Grease, 14 Ounce (For Cartridge-Type Grease Gun). |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-G-81322 | - Grease, Aircraft, General Purpose, Wide Temperature Range. |



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STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw Thread Standards for Federal Service. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-889 | - Dissimilar Metals. |

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hand-operated, lever, push, and screw type grease guns (hereinafter called "grease guns"), covered by this specification shall be contractor's current standard commercial product except as specified herein. Except for materials and part numbering system, MIL-G-3859/1 through MIL-G-3859/4 are for identification purposes only, and are not intended to preclude grease guns, otherwise conforming to this specification (see 6.4).



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3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of grease guns from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish grease guns conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The grease guns shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the grease guns may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the grease gun may be newly fabricated from recovered materials to the maximum extent practicable, provided the grease gun produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the grease gun.

3.4 Design and construction. Grease guns covered by this specification shall conform to the description and requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for replacement. The grease cylinder or barrel shall be of metal, or durable and petroleum product resistant plastic, of ample strength so that the gun will not break or become distorted when it is dropped on a concrete floor from a height of 48 inches (see 4.5.2.1).

3.5 Threads. All threaded parts shall comply with FED-STD-H28.

3.6 Fastening devices. All fastening devices shall be installed with positive means for preventing them from loosening or becoming out of adjustment. All fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.



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3.7 Castings and forgings. All castings and forgings, when used in construction of the gun shall be free from blowholes, porosity, shrinkage defects, and cracks. Forgings shall be free from scale, inclosures, cold shuts, or mismatching. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. In no event shall welding, brazing, soldering, peening or plugging be used on castings or forgings for reclaiming any parts of the grease gun without authorization of the contracting officer.

3.8 Finish. All exposed metal surfaces shall be corrosion-resistant painted or plated to withstand corrosion.

3.9 Gripping surfaces. Non-slip hand gripping surfaces shall be provided wherever gripping or assembling and disassembling by hand is required.

3.10 Identification marking. Grease guns shall be stamped for identification with a metal die in accordance with standard commercial practice.

3.11 Performance.

3.11.1 Storage conditions. The grease gun and all components thereof shall be capable of withstanding storage at any temperature between -25° Fahrenheit (F) and +160 °F and at any relative humidity condition between 0 and 100 percent without sustaining damage such as the gun or its components buckling, twisting, tearing, or fracturing.

3.12 Type I grease gun. The type I grease gun shall be lever operated and similar to figure 1 of MIL-G-3859/1 designed to load with a grease cartridge or bulk loading (by suction or paddling). When specified (see 6.2), a straight fitting conforming to MIL-F-3541, type III shall be installed in the head to enable pump loading through hydraulic couplings. When this fitting is specified, the following note shall be permanently placed as close to the fitting as possible: "TO FILL - OPEN HANDLE".

3.12.1 Details of components. Each grease gun shall consist essentially of a head, lever assembly, barrel, follower assembly, with an extension and coupling assembly.

3.12.1.1 Head and lever assembly. The head shall consist of a casting or shall be fabricated from welded steel construction incorporating a positive check valve. The head shall rigidly support the operating lever linkage. The discharge outlet thread shall be tapped 1/8-27 NPTF. The cylinder shall be of ream or similar finish without scratches or bell mouth. A sealing gasket of durable oil-proof material shall be provided to prevent leakage of lubricant at the barrel and head connection. This gasket shall be self-retained by the head when removed from the barrel. The head shall incorporate an internal thread to match that of the barrel and a suitable counterbore for pilot action when connecting the barrel to the head. The head gasket and other gun sealing members shall retain the lubricant under all specified operating conditions.



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3.12.1.2 Barrel and follower assembly. The barrel shall be steel tubing with a wall thickness of not less than 0.044 inch or a material equivalent in all respects, approximately 2.25 inches outside diameter in size 14. The barrel shall have an outside thread at one end for mounting to the head assembly. Barrels which have different size threads or different number of threads on either end shall be stamped with a marking "NOZZLE END" and an " ⟶ " (ARROW) pointing to the nozzle end. The barrel shall have an effective nonslip hand-gripping surface located in the normal hand-gripping area. The follower assembly shall consist of a plunger, a helical follower spring having a maximum thrust of 40 pounds when fully compressed (see 4.5.2.4), and a follower rod securely fitted with a handle. The plunger shall not allow any grease to pass around it when grease is being compressed by the follower assembly. A suitable latch device that can be operated under all environmental conditions specified herein shall be provided to engage and hold the follower rod when the follower spring is compressed for loading grease or grease cartridges conforming to MIL-C-22587.

3.12.1.3 Extension and coupling assembly. An extension and coupling assembly conforming to MIL-L-4387, type III, class 1 (bent), shall be furnished with each gun, except that size 1 coupling shall be furnished in lieu of the size 2 coupling. Couplers with nonreversible jaws which otherwise conform to MIL-L-4387, type I, class I, style A, may be furnished.

3.12.1.4 Minimum pressure for type I grease gun. Type I grease gun shall develop a minimum pressure of 6000 pounds per square inch gage (psig) when tested in accordance with 4.5.2.5. There shall be no evidence of leakage or physical damage to the grease gun at any time during or after this test.

3.12.1.5 Dispensing rate type I and type IV grease gun. When filled with any one of the lubricants as specified in MIL-G-10924, MIL-G-23827, and MIL-G-81322 and when normally lever operated at a temperature of between 50 °F to 75 °F, the type I and type IV, size 14, size 18 and size 80 grease gun shall dispense not less than 1 ounce of grease (by weight) per each 40 strokes (see 4.5.2.6). The type IV, size 03 grease guns shall dispense not less than 0.5 ounce of grease (by weight) per each 40 strokes (see 4.5.2.6).

3.12.1.6 Back pressure effect - types I and IV grease guns. When subjected to a minimum back pressure of 3000 psig, type I and type IV, size 14, size 18, and size 80 grease guns, filled and operated as specified in 3.12.1.5, shall dispense not less than 0.8 ounce of grease (by weight) per each 40 strokes. Type IV, size 03 grease guns, when filled and operated as specified above, shall dispense not less than 0.4 ounce of grease (by weight) per each 40 strokes (see 4.5.2.7).

3.13 Type II grease gun. The type II grease gun shall be of the straight type with the hydraulic coupling, compressor, and follower rod in a straight line (see figure 1 of MIL-G-3859/2). The end cap shall be provided with either a knurled, serrated or ribbed grip, except the class B gun which shall be of the "K" or angle type for one hand operation (see figure 2 of MIL-G-3859/2). The container for the grease shall be the handle portion of the gun. The



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compressor, push knob, and hydraulic coupling shall be in a straight line. The handle portion and the end cap shall be either knurled, serrated or ribbed to provide ease of handling and disassembly for loading purposes.

3.13.1 Loader fitting. When specified (see 6.2) the type II grease gun shall be supplied with a loader fitting conforming to MIL-F-3541, type III.

3.13.2 Hydraulic coupling. The hydraulic coupling shall conform to MIL-L-4387, type I, class 1, style A, size 1. For the class B grease gun this coupling shall be connected through an extension. The extension shall conform to type III straight extension of MIL-L-4387, except that the length shall be 1.25 inches minimum (including the length of threads).

3.13.3 Primer feed plunger. The primer feed plunger shall be actuated by means of spring or hand push rod and shall feed grease into the compressor without leakage. After priming, the grease shall feed automatically into the compressor without further priming. Materials used shall not be adversely affected by lubricants.

3.13.4 Compressor. The type II, size 03, class A and class B compressor shall be of the piston type, and shall be capable of withstanding pressures up to 4000 psig. The type II, size 10 and size 28 compressor shall withstand 4300 psig when force is applied to the handle or push knob as applicable. The compressor chamber shall refill itself when the handle or push knob is returned to its normal position.

3.13.5 Push handle. The handle of the type II grease gun shall be shaped to form a natural hand grip.

3.13.6 Push knob. The push knob of the type II, size 03, class B grease gun shall be a smooth saucer-shaped steel disk, with the convex surface shaped for hand pushing. The knob shall retract to its normal position by spring action.

3.13.7 Dispensing capacity of type II grease gun. The type II grease gun shall perform as specified herein after being stored at -25 °F for not less than 8 hours with no solar radiation. When filled with grease conforming to either MIL-G-10924 or MIL-G-23827, the type II, size 03, class A gun shall dispense not less than 2.5 ounces (by weight) of grease at a rate not less than 0.75 ounce of grease in 150 strokes at a temperature of 70 °F ±5 °F. When filled as specified above, the type II, size 03, class B grease gun shall dispense not less than 2.5 ounces (by weight) of grease at a rate of not less than 0.50 ounce of grease in 150 strokes under the same temperature condition (see 4.5.2.8). The gun shall show no cracks, fractures or permanent deformation as a result of the low temperature storage test.

3.13.8 Static pressure. The type II, size 03, class A and B grease gun shall develop an internal pressure of 4000 psig, and the type II, size 10 and 28 grease gun shall develop 4300 psig, without any evidence of leakage or distortion (see 4.5.2.9).



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3.13.9 Delivery pressure. The type II grease gun shall deliver not less than 0.24 ounces of grease in 50 strokes through the coupling against a back pressure of 3000 psig with the coupling at any angle up to and including 10 degrees from the axis of any fitting conforming to MIL-F-3541 (see 4.5.2.10).

3.14 Type III grease guns. The type III 8-ounce capacity, screw operated, hand grease gun shall be similar to figures 1 and 2 of MIL-G-3859/3. Each grease gun shall consist essentially of a head, barrel, and follower assembly.

3.14.1 Type III, outlet A grease gun. The type III, outlet A grease gun shall have an overall length of 10 to 14 inches when the handle is in its outermost position, and be similar to figure 1 of MIL-G-3859/3. The grease outlet shall be a female 1/8-27 NPTF, and shall be furnished with an installed pipe plug (1/8-27 NPTF).

3.14.1.1 Attachment. When specified (see 6.2), a flexible rubber hose assembly, rigid bent angle tube assembly, and an adapter shall be furnished (see figure 1 of MIL-G-3859/3).

3.14.1.2 Flexible rubber hose assembly. The flexible rubber hose assembly shall conform to MIL-H-13444, type I or II, 1/4-inch nominal inside diameter. The fittings shall be furnished with 1/2-20 Unified National Fine (UNF) threads. The male fitting shall have a 45 degree chamfer as indicated in figure 1 of MIL-G-3859/3 and the female fitting shall be of the 45 degree flared type. The length over the fittings shall be not less than 10.5 inches.

3.14.1.3 Rigid bent angle tube. The rigid bent angle tube shall be 1/8-inch inside diameter by 3/16-inch outside diameter. The nonmating end shall be bent approximately 25 degrees at a distance of 1 inch from the tip. The tube shall have a steel or brass female fitting silver soldered or copper brazed to the straight end as shown in figure 1 of MIL-G-3859/3. The fitting shall be of the 45-degree flared type with a 1/2-20 UNF female thread. The overall length of this assembly shall be no less than 4.25 inches.

3.14.1.4 Adapter. A male adapter shall be provided with a 1/2-20 UNF thread on one end, and a 1/8-27 NPTF thread on the other end. The adapter shall have a 45 degree chamfer on the 1/2-inch end, as indicated in figure 1 of MIL-G-3859/3.

3.14.2 Type III, outlet B grease gun. The type III, outlet B grease gun shall have an overall length of 16 to 20 inches with the handle in its outermost position, and be similar to figure 2 of MIL-G-3859/3. The grease outlet shall be such that it will securely mate with a standard "dot" fitting, by means of a standard "dot" coupling which shall be furnished as a component part of each gun.

3.14.2.1 Bottom cone. The bottom cone shall have internal threads to fit the end of the cylinder and external threads to mate with the nozzle and valve assembly (standard "dot" coupling). The exterior of the cone shall be knurled, as indicated in figure 2 of MIL-G-3859/3, to provide satisfactory gripping with the hand for removal and assembly.

3.14.3 Cylinder. The grease cylinder for type III grease gun shall allow for satisfactory gripping with the hand by means of knurling, serrations or ribbing



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on the outside diameter to a minimum length of 3.5 inches, located at the discharge end. The inside diameter shall be round and smooth; and the cap end shall be chamfered to facilitate entrance of the leather piston.

3.14.4 Handle. The handle for type III grease gun shall consist of a bar no less than 3 inches long, of convenient shape for hand operation, centered and securely fixed to the projecting end of the screw so as to form a "T".

3.14.5 Cap. The cap for type III grease gun shall be dome-shaped and threaded to fit the end of the cylinder. A threaded opening shall be provided for a fit with the screw. The exterior shall be knurled, as indicated in figures 1 and 2 of MIL-G-3859/3 to provide satisfactory gripping with the hand for removal and assembly. The interior of the cap shall permit the piston and follower to be drawn completely within the cap. In this position the leather piston shall be enclosed, protected, and compressed to facilitate entry into the cylinder.

3.14.6 Dispensing capacity type III grease gun. The type III grease guns shall dispense not less than 6 ounces (by weight) of grease at temperatures ranging from -25 °F to +125 °F when tested as specified in 4.5.2.11 and 4.5.2.12.

3.14.7 Back pressure type III grease gun. The type III grease gun shall deliver grease against a back pressure of 500 psig, and shall equalize grease against a back pressure of 700 psig, without evidence of leakage or damage when tested as specified in 4.5.2.13.

3.15 Type IV grease guns. The type IV pistol operated grease guns shall be similar to figure 1 of MIL-G-3859/4. The grease gun shall consist essentially of a head, barrel, and follower assembly constructed to load with a cartridge and bulk loading. When specified (see 6.2), for size 14, a straight fitting conforming to MIL-F-3541, type III shall be installed in the head to enable pump loading through a hydraulic coupling. When this fitting is specified (see 6.2), the following note shall be permanently placed as close to the fitting as possible: "TO FILL - OPEN HANDLE".

3.15.1 Details of components. The grease gun shall consist of a head pistol grip assembly barrel follower assembly, with an extension and coupling assembly.

3.15.2 Head and pistol grip assembly. The head shall consist of a casting or shall be fabricated from welded steel construction incorporating a positive check valve. The pistol grip type lever shall be attached to the head with a linkage to apply force to the end of the piston. The discharge outlet thread shall be tapped 1/8-27 NPTF. The sealing gasket of durable oil proof material shall be provided to prevent leakage of lubricant at the barrel and head connection. This gasket shall incorporate an internal thread to match that of the barrel and a suitable counterbore for pilot action when connecting the barrel to the head. The head gasket and other gun sealing members shall retain the lubricant under specified operating conditions.

3.15.3 Barrel and follower assembly. The barrel and follower assembly shall be as specified in 3.12.1.2.



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3.15.4 Extension and coupling assembly. The extension assembly shall conform to MIL-L-4387, type IV, class 1, style C. The coupling shall conform to MIL-L-4387, type I, class 1, style A or type I, class 4. Couplers with nonreversible jaws which otherwise conform to MIL-L-4387, type I, class 1, style A may be furnished.

3.15.5 Minimum pressure. The type IV grease gun shall have a pressure of 6000 psi when tested as specified in 4.5.2.5. There shall be no evidence of leakage or damage to the grease gun at any time during this test.

3.15.6 Dispensing rate. The type IV grease gun dispensing rate shall be as specified in 3.12.1.5.

3.15.7 Back pressure. The type IV grease gun back pressure shall be as specified in 3.12.1.6.

3.16 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.17 Workmanship. All parts, components, and assemblies of the grease gun including casting, forgings, molded parts, stampings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.17.1 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed only in accordance with commercial practice. These operations shall not be employed as repair measures. In no event shall such processes, as peening or plugging, be used on castings or forgings for reclaiming any parts of the grease gun.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The



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absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article grease gun shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article grease gun shall be tested as specified in 4.5.2.1 through 4.5.2.13, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-3.

4.4.1.1 Inspection lot. A lot shall consist of all grease guns of one type submitted for inspection at one time. The sample unit shall be one completely fabricated grease gun.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Acceptable Quality Level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.5 through 4.5.2.10 and 4.5.2.13 as applicable. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The grease guns shall be examined as specified herein, for the following defects:



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Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the grease guns.
106. Parts subject to wear inaccessible for adjustment.
107. Threads not provided where specified.
108. Grease gun inoperative, or abnormal force required to operate it.
109. Insecure fastening devices.
110. Burrs and sharp edges not removed.
111. Omission of a component part.
112. Capacity of container of the gun not as specified.
113. Angle and fit of accessories not as specified.
114. Knurling, serrations, or ribbing not provided as specified.
115. Castings and forgings not as specified.
116. Dimensions not as specified.
117. Gripping surfaces not as specified.
118. Welding, brazing, and soldering not as specified.
119. Type I grease gun(s) not as specified.
120. Head and lever assembly not as specified.
121. Extension and coupling assembly not as specified.
122. Type II grease gun(s) not as specified.
123. Loader fitting not as specified.
124. Hydraulic coupling not as specified.
125. Primer feed plunger not as specified.
126. Compressor not as specified.
127. Push handle not as specified.
128. Push knob not as specified.
129. Type III grease gun(s) not as specified.
130. Attachment not as specified.
131. Type III, outlet A grease gun not as specified.
132. Flexible rubber hose assembly not as specified.
133. Rigid bent angle tube not as specified.
134. Adapter not as specified.
135. Type III outlet B grease gun not as specified.
136. Bottom cone not as specified.
137. Cylinder not as specified.
138. Handle not as specified.
139. Cap not as specified.
140. Type IV grease gun(s) not as specified.
141. Barrel and follower assembly not as specified.
142. Extension and coupling assembly not as specified.
143. Head and pistol grip assembly not as specified.
144. Interchangeability not as specified.



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Minor

201. Identification marking missing, incomplete, illegible or not permanent.
202. Blistered, peeled, chipped, scratched, foreign matter embedded or not uniform.

4.5.2 Tests.

4.5.2.1 Drop test. The empty grease gun shall be dropped three times from a height of not less than 48 inches onto a concrete floor. The test shall be repeated with the gun filled to maximum capacity. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.2 Storage conditions. The grease gun shall be conditioned at a temperature of -25 ±5 °F with humidity for a period of 24 hours, and then equalized at room temperature (65 °F to 75 °F). If there is no evidence of damage, the grease gun shall then be subjected to 5 hours of steady increase to +160 ±5 °F, be held at this temperature for 4 hours, and equalized at room temperature (65 °F to 75 °F). The grease gun shall then be inspected for compliance with 3.11.1. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.3 Humidity test. The grease gun shall be tested in accordance with the following procedure for 2 cycles (48 hours) and then be reinspected for compliance with 3.11.1. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.3.1 Procedure. The equipment shall be placed in the test chamber and set up to simulate installed conditions. The test chamber shall be vented to the atmosphere to prevent the build up of pressure. Prior the the starting of the test period, the chamber temperature shall be between 20 °C and 38 °C (68 °F and 100 °F) with uncontrolled humidity. During the first 2-hour period, the temperature shall be gradually raised to 11 °C (160 °F). The temperature selected shall be maintained during the next 6-hour period. The velocity of the air throughout the test area shall not exceed 150 feet per minute. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20 °C to 38 °C (68 °F to 100 °F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. At the conclusion of 2 cycles, the 48 hour period, the grease gun shall be returned to standard conditions.

4.5.2.4 Follower spring test. The follower spring shall be fully compressed and by means of tension gauge or other means, be examined for compliance with 3.12.1.2. Failure to comply with 3.12.1.2 shall constitute failure of this test.

4.5.2.5 Minimum pressure test type I and type IV grease guns. The type I and type IV guns with extension and coupling attached shall be loaded with grease conforming to MIL-G-81322 or MIL-G-10924. The gun shall be operated to precondition the pressure measuring apparatus to 6000 psig for the type I and type IV gun. By means of a pressure relief valve the system pressure shall be adjusted to a minimum pressure of 4000 psig. The relief valve shall be closed during the test to allow the gun to attain the specified minimum pressure. The gun shall then be operated by applying a force to the end of the piston in the



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direction of the piston centerline. This test shall be repeated ten times and the gun shall be examined for leakage or physical damage. Nonconformance to 3.12.1.4 shall constitute failure of this test.

4.5.2.6 Dispensing rate type I and type IV grease gun. The type I and type IV grease gun shall be operated for a minimum of 40 strokes, dispensing the grease into a preweighed container to determine conformance to 3.12.1.5. Nonconformance to 3.12.1.5 shall constitute failure of this test.

4.5.2.7 Back pressure effect type I and type IV grease gun. The type I and type IV grease guns shall be attached to the same or similar apparatus used for the minimum pressure test, except that a pressure relief valve designed to open at 3000 psig shall be installed. The grease gun shall then be operated for a minimum of 40 strokes, dispensing the grease into a preweighed container to determine conformance to 3.12.1.6. Nonconformance to 3.12.1.6 shall constitute failure of this test.

4.5.2.8 Dispensing type II grease gun. The type II grease gun shall be filled with grease conforming to either MIL-G-10924 or MIL-G-23827 and stored at -25 ±5 °F for a period of 8 hours with no solar radiation, then removed and allowed to equalize at 70 ±5 °F. The grease gun shall be operated in the normal manner to dispense the grease into a preweighed container. When the gun will no longer dispense grease, the net weight of the dispensed grease and condition of the gun shall be examined for compliance with 3.13.7. Nonconformance to 3.13.7 shall constitute failure of this test.

4.5.2.9 Static pressure (leakage) test. The type II, size 1, class A and B grease gun shall be coupled to a lubrication fitting and pressure gauge, and then operated until a gauge pressure of 4000 psig has been reached, and 4300 psig for the type II, size 2 and 3 grease gun. This procedure shall be repeated ten times, and the grease gun observed for conformance to 3.13.8. Nonconformance to 3.13.8 shall constitute failure of this test.

4.5.2.10 Delivery pressure. The type II grease gun shall be filled with the same type of grease used previously (see 4.5.2.8). Operate the grease gun for not less than 50 strokes against a pressure of at least 3000 psig, and at an angle up to and including 10 degrees from the axis of the fitting. Measure the amount of grease dispensed by the gun. Nonconformance to 3.13.9 shall constitute failure of this test.

4.5.2.11 Cold test. The type III outlet A and B grease gun shall be filled with grease conforming to MIL-G-10924, and allowed to equalize at -25 ±5 °F. The screw shall then be manually turned into the cylinder as far as possible, dispensing the grease into a preweighed container. The net weight of the dispensed grease shall be determined. Nonconformance to 3.14.6 shall constitute failure of this test.

4.5.2.12 Storage test. Repeat the test outlined in 4.5.2.11, except that the temperature shall be equalized at +125 ±5 °F. Nonconformance to 3.14.6 shall constitute failure of this test.



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4.5.2.13 Back pressure and leakage test type III grease gun. After completion of the above tests, refill the grease gun with the same type grease previously used, and equalize at a temperature of +125 ±5 °F. The type III outlet A grease gun shall be connected directly to a pressure gauge and the type III outlet B grease gun shall be connected to a "dot" type lubrication fitting, connected to the pressure gauge. The gauge shall have a dial reading of no less than 1000 psig. The back pressure shall then be adjusted to 500 psig. The grease gun shall be hand operated and shall discharge grease against the above specified back pressure. In addition, upon completion of this test, the screw shall be operated so as to increase the gauge reading to 700 psig. Without turning the screw any further, the grease gun shall be held in this position for 3 minutes. The total pressure drop within this time lapse shall not exceed 50 psig. Nonconformance to 3.14.7 shall constitute failure of the test.

4.5.2.14 Test for repair parts. All recommended repair parts which are subject to wear, breakage or distortion shall be tested for ease of removal and replacement. Nonconformance shall constitute failure of the test.

4.6 Inspection of packaging. The packaging of the grease guns shall be examined and tested in accordance with the applicable preservation, packaging, packing, and marking quality assurance provisions of PPP-P-40.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2), in accordance with the preservation and packaging requirements of PPP-P-40 except that MIL-STD-1188, which is referenced therein, shall be replaced by ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2), in accordance with the packing requirements of PPP-P-40 except that MIL-STD-1188, which is referenced therein, shall be replaced by ASTM D 3951.

5.3 Marking. All containers shall be marked for all degrees of packaging in accordance with the applicable marking requirements of PPP-P-40. Commercial marking shall be in accordance with ASTM D 3951 rather than MIL-STD-1188. Additionally, each shipping container shall be marked with the cube and gross weight.

6. NOTES

6.1 Intended use. The grease guns covered by this specification are intended for use by military maintenance organizations in delivering grease at low and medium pressure, and also for general and special purpose high-pressure greasing operations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, size, class of outlet grease gun required (see 1.2).
c. Specification based part number required (see 1.3).
d. When a first article is required and number of grease guns required (see 3.2).



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e. If fitting for pump loading is required (see 3.12, 3.13.1 and 3.15).
f. When flexible rubber hose assembly, rigid bent angle tube assembly and an adapter is required (see 3.14.1.1).
g. When a straight fitting conforming to MIL-F-3541, type III is required (see 3.15).
h. Degree of preservation and degree of packing required (see 5.1 and 5.2 respectively).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Information figures. MIL-G-3859/1 through MIL-G-3859/4 figures show grease guns which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the purchase of other grease guns which conform to this specification.

6.5 Subject term (key word) listing.

Grease gun, hand operated
Grease gun, lever operated
Grease gun, push type
Grease gun, screw type
Gun, grease, hand operated

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Navy - SH
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User activities:
Army - CE
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Preparing activity:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-G-3859E | Grease Guns, Hand-Operated, Lever, Push and Screw Type General Specification For |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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